

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full stack development MERN Practical

Practical – 2

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Class: TYIT	Batch: 1
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Aim:-

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code:-

```
function calculateSalary(name,basicSalary){
    let bonus = 5000;
    let totalSalary = basicSalary + bonus;

    console.log("Employee Name ",name);
    console.log("Employee Salary ",basicSalary);
    console.log("Employee Bonus",bonus);
    console.log("total Salary ",totalSalary);
}
calculateSalary("Aashu Yadav", 40000);
console.log("Aashu ")
```

```
function generateBill(product,quantity , price){
    let total = 5000;
    let Bill = total + (total*0.09);

    console.log("product Name ",product);
    console.log("product quantity ",quantity);
    console.log("price par item ",price);
    console.log("total Bill",Bill);
}
generateBill("phone",2, 50000);
console.log("Aashu ")
```

Output:-

```
PS C:\Users\itlab\Desktop\TYIT Aashu MERN> node pract2A.js
Employee Name  Aashu Yadav
Employee Salary  40000
Employee Bonus  5000
total Salary  45000
Aashu
```

```

PS C:\Users\itlab\Desktop\TYIT Aashu MERN> node pract2A.js
product Name  phone
product quantity  2
price par item  50000
total Bill  5450
Aashu

```

b. Work with arrays and objects in JavaScript.

Code:-

```

let fruits=["Mango","Apple","cherry","banana"];
let student={
  name:"Aashu ",
  age: 20,
  city:"Mumbai"
};
console.log("fruit",fruits);
console.log(" Second fruit",fruits[1]);
console.log("student Name :",student.name);
console.log("student Age:",student.age);
console.log("student city:",student.city);
console.log("Aashu");

```

Output:-

```

PS C:\Users\itlab\Desktop\TYIT Aashu MERN> node pract2B.js
fruit [ 'Mango', 'Apple', 'cherry', 'banana' ]
Second fruit Apple
student Name : Aashu
student Age: 20
student city: Mumbai
Aashu

```

c. Demonstrate use of arrow functions.

Code:-

```

let add =(a, b) => {
  return a + b;
};

let sub =(a, b) => {
  return a - b;
};

let mul =(a, b) => {
  return a * b;
};
console.log("Sum=",add(10,20));
console.log("subtraction=",add(30,20));
console.log("multiply=",add(10,10));
console.log("Aashu");

```

Output:-

```

PS C:\Users\itlab\Desktop\TYIT Aashu MERN> node pract2C.js
Sum= 30
subtraction= 50
multiply= 20
Aashu

```

d. Write a program using array methods (map, filter, reduce).

Code:-

```
let numbers =[10,20,30,40,50];
let doubled =numbers.map(num => num *2);
let greater25 = numbers.filter(num => num > 25);
let total =numbers.reduce((sum, num) => sum + num ,0);

console.log("original Array:",numbers);
console.log("map:",doubled);
console.log("Filter:",greater25);
console.log("reduce:",total);
console.log("Aashu");
```

Output:-

```
PS C:\Users\itlab\Desktop\TYIT Aashu MERN> node pract2D.js
original Array: [ 10, 20, 30, 40, 50 ]
map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
reduce: 150
Aashu
```

e. Create a program that manipulates objects and displays output.

Code:-

```
let student ={
  rollNo: 50,
  Name:"Aashu ",
  marks:86
};
console.log("original object:");
console.log(student);

//accessing properties
console.log("/nAccessing properties:");
console.log("name",student.Name);
console.log("Marks",student.marks);

// updating a property
student.marks=90;
console.log("/nAfter updating marks:");
console.log(student);

// adding a new property
student.city ="mumbai";
console.log("/n adding a new property:");
console.log(student);

//delete a property
delete student.rollNo;
console.log("/ndelete a property");
console.log(student);
console.log("Aashu");
```

Output:-

```
PS C:\Users\itlab\Desktop\TYIT Aashu MERN> node pract2E.js
original object:
{ rollNo: 50, Name: 'Aashu ', marks: 86 }
/nAccessing properties:
name Aashu
Marks 86
/nAfter updating marks:
{ rollNo: 50, Name: 'Aashu ', marks: 90 }

adding a new property:
{ rollNo: 50, Name: 'Aashu ', marks: 90, city: 'mumbai' }

delete a property
{ Name: 'Aashu ', marks: 90, city: 'mumbai' }
Aashu
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code:-

```
let students =[
  {
    rollNo: 50,
    name: "Aashu",
    marks: 85
  },
  {
    rollNo: 17,
    name: "Radhika",
    marks: 90
  },
  {
    rollNo: 2,
    name: "Asmitha",
    marks: 80
  }
];
function display(s)
{
  console.log("Student details");
  console.log("-----");
  for (let stud of s)
  {
    console.log(`Roll no : ${stud.rollNo}`);
    console.log(` name : ${stud.name}`);
    console.log(`Marks : ${stud.marks}`);
    console.log(`-----`);
  }
}
display(students);
console.log("Aashu");
```

output:-

```
PS C:\Users\itlab\Desktop\TYIT Aashu MERN> node pract2F.js
Student details
-----
Roll no : 50
name : Aashu
Marks : 85
-----
Roll no : 17
name : Radhika
Marks : 90
-----
Roll no : 2
name : Asmitha
Marks : 80
-----
Aashu
```